

DSD Final Project Scores

1. BrPred

(1) Total execution cycles of given I_mem_BrPred: 558

截圖:

```
[INFO]: Testing with pattern <BrPred>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsvd/CBDBK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in F
inal_tb.chip0.core0_ex_stage0_mul_count_reg_1_
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-72) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:          558
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time        1507647
      V C S   S i m u l a t i o n   R e p o r t
Time: 1507647 ps
```

(2) Total execution cycles of given I_mem_hasHazard: 2375

截圖:

```
[INFO]: Testing with pattern <hasHazard>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsvd/CBDBK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in F
inal_tb.chip0.core0_ex_stage0_mul_count_reg_1_
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-72) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:          2375
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time        6413547
      V C S   S i m u l a t i o n   R e p o r t
Time: 6413547 ps
```

(3) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): 2.7 (ns)

2. Compressed instructions

(1) Total Simulation Time of given I_mem_compression: 1874.847 (ns)

截圖：

```
[INFO]: Testing with pattern <compression>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsg/CBDK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in F
inal tb.chip0.core0_ex_stage0_mul_count_reg_1
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-72) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:          694
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time          1874847
```

(2) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): 2.7 (ns)

3. Multiplication instructions

(1) Total Simulation Time of given Mul/I_mem: 738.147 (ns)

截圖：

```
[INFO]: Testing with pattern <Mul>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsg/CBDK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in F
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-72) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:          273
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time          738147
V C S   S i m u l a t i o n   R e p o r t
Time: 738147 ps
```

(2) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): 2.7 (ns)

4. Q_sort & Conv & LFSR_HIST

(1) Area: 769950.832599 (um²)

截圖:

```
Current design is 'CHIP'.
CHIP
dc_shell> report_area

*****
Report : area
Design : CHIP
Version: U-2022.12
Date   : Wed Jun 10 15:04:54 2026
*****

Information: Updating design information... (UID-85)
Warning: Design 'CHIP' contains 1 high-fanout nets. A fanout number of 1000 will be used for delay calculations involving these nets. (TIM-134)
Library(s) Used:

    typical (File: /home/raid7_2/course/cvstd/CBDK_IC_Contest/CIC/SynopsysDC/db/typical.db)

Number of ports:                577
Number of nets:                 44893
Number of cells:                40022
Number of combinational cells:  28294
Number of sequential cells:     11595
Number of macros/black boxes:   0
Number of buf/inv:              4270
Number of references:           216

Combinational area:             410288.743927
Buf/Inv area:                   43202.225287
Noncombinational area:          359662.088673
Macro/Black Box area:           0.000000
Net Interconnect area:          5322314.016083

Total cell area:                769950.832599
Total area:                     6092264.848682
1
dc_shell> █
```

(2) Total Simulation Time of Q_Sort: 193798.947 (ns)

(either using compressed or uncompressed instructions) uncompressed instructions

截圖:

```
[INFO]: Testing with pattern <QSort_uncompressed>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvstd/CBDK_IC_Contest/CIC/Verilog/tsmc13.v", 23104: Timing violation in Final_tb.chip0.core0_ex_stage0_mul_count_reg_1_
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-72) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:          71777
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time        193798947
```

(3) Total Simulation Time of Conv: 34423.347 (ns)

(either using compressed or uncompressed instructions) uncompressed instructions

截圖:

```
[INFO]: Testing with pattern <Conv_uncompressed>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsc/CBDK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in Final_tb.chip0.core0_ex_stage0_mul_count_reg_1_
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-7
2) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:      12749
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time    34423347
      V C S   S i m u l a t i o n   R e p o r t
Time: 34423347 ps
```

(4) Total Simulation Time of LFSR_HIST: 722164.647 (ns)

截圖:

```
[INFO]: Testing with pattern <LFSR_HIST>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====

"/home/raid7_2/course/cvsc/CBDK_IC_Constest/CIC/Verilog/tsmc13.v", 23104: Timing violation in Final_tb.chip0.core0_ex_stage0_mul_count_reg_1_
$setuphold( posedge CK &&& (flag == 1'b1):5400, negedge D:5288, limits: (115,-7
2) );

===== Reset done =====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
Total execution cycles:      267468
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 200.
$finish at simulation time    722164647
      V C S   S i m u l a t i o n   R e p o r t
Time: 722164647 ps
```

(5) $A \times T_{Q_sort} \times T_{Conv} \times T_{LFSR_HIST} : 3.7094 \times 10^{21} (um^2 * ns^3)$

(6) Clock cycle for post-syn simulation (cycle in testbench, not cycle in sdc): 2.7 (ns)